ATLANTIS: An Anatomy-guided Tissue-Latent Atlas Network for DWI-derived White-Matter Anisotropy Synthesis from T1w MRI

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***Abstract*—Fractional anisotropy (FA), estimated by diffusion tensor modeling of multi-directional diffusion-weighted imaging (DWI), is a representative white-matter anisotropy biomarker for quantifying diffusion directionality in white-matter microstructure. However, diffusion acquisition is time-consuming and often unavailable in retrospective or routine clinical datasets, whereas T1-weighted (T1w) structural MRI is acquired in most neuroimaging protocols. While recent deep learning methods have demonstrated the feasibility of synthesizing FA maps from T1w MRI, evaluation has largely focused on voxel-level similarity metrics, leaving tract-level FA biomarker preservation underexplored. We propose ATLANTIS, an Anatomy-guided Tissue-Latent Atlas Network instantiated for FA map synthesis from T1w MRI. ATLANTIS integrates tissue-aware bottleneck modulation, prior-restoration decoding with multi-scale anatomy skip connections, and voxel-wise adaptive fusion. On the HCP Young Adult dataset, ATLANTIS achieves an image-level MAE of 0.0402 (5.0% lower than ResViT) and substantially larger gains at the tract level: average tract MAE of 0.0067 (20.2% reduction) and average absolute bias of 0.0008 (83.3% reduction) across eight JHU white-matter tracts. Ablation identifies the Restoration Decoder as the dominant contributor, with its removal increasing MAE by 35.3%. These results indicate that anatomy-guided synthesis benefits tract-level biomarker preservation beyond voxel-level metrics, supporting the feasibility of diffusion-informed FA analysis from structural MRI.**

***Index Terms*—** DWI-derived white-matter anisotropy, fractional anisotropy, T1w MRI, structural-to-diffusion MRI synthesis, tract-level biomarker preservation, medical image synthesis

I. Introduction

Fractional anisotropy (FA), derived by diffusion tensor modeling of multi-directional diffusion-weighted imaging (DWI), is widely used as a quantitative index associated with white-matter microstructural organization [1], [2]. Although FA is not specific to a single biological substrate, tract-mean FA values from atlas-defined white-matter regions are widely used in population neuroscience and clinical research. However, high-quality diffusion acquisition is time-consuming and may be unavailable in retrospective studies, routine clinical cohorts, or datasets acquired with structural MRI protocols only [3]. In contrast, T1-weighted (T1w) structural MRI is among the most routinely acquired neuroimaging sequences [4]. Synthesizing FA maps from T1w MRI could therefore extend diffusion-informed white-matter analysis to cohorts without dedicated DWI/DTI acquisition.

Recent advances in medical image synthesis have enabled the estimation of missing or alternative imaging modalities, with reviews summarizing progress across CNN-, Transformer-, and diffusion-based models for MRI, CT, and PET synthesis [5]. GAN-based architectures such as Pix_2Pix [6] provided early demonstrations of paired image translation, while CNN-based methods including 3D U-Net [7] extended this capability to volumetric data. Transformer-based approaches such as ResViT [8] and diffusion-bridge methods such as I^2SB [9] have further expanded the methodological toolkit. These architectures form the baseline categories evaluated in this work. Several studies have specifically targeted T1w-to-FA or T1w-to-diffusion synthesis. Gu et al. [3] used CycleGAN to generate FA and mean diffusivity maps from T1w images. Anctil-Robitaille et al. [10] proposed a manifold-aware CycleGAN for structural-to-DTI tensor synthesis. Murcia Tapias et al. [11] demonstrated T1w-to-FA synthesis using a simplified GAN, and Macro2Micro [12] recently explored multi-scale brain-structure encoding for T1w-to-FA translation. More recent work has further investigated transfer-learning-based T1-to-FA synthesis for white-matter assessment in stroke cohorts [13], highlighting growing interest in deriving diffusion-informed biomarkers from structural MRI. These studies establish the feasibility of structural-to-diffusion scalar-map or tensor prediction. Yet evaluation in these works is generally centered on image-level or voxel-level reconstruction quality, and image-level fidelity does not necessarily guarantee preservation of biomarker-level quantities. More generally, reference-based image similarity metrics may fail to reflect downstream or biomarker-level utility in medical image synthesis [14]. For synthetic FA, this issue is particularly

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important because downstream analyses often rely on tract-mean FA rather than individual voxel intensities. A synthetic FA map with low global error may still introduce systematic bias in tract-level FA estimates, potentially affecting biomarker interpretation in group comparisons or individual patient assessments. Systematic atlas-based assessment of tract-level FA preservation therefore remains underexplored.

We propose ATLANTIS, an anatomy-guided tissue-latent atlas network for DWI-derived white-matter anisotropy synthesis from T1w MRI. In this work, we instantiate this framework by predicting FA maps, a representative DWI/DTI-derived anisotropy biomarker. Our contributions are as follows. First, we introduce an anatomy-guided synthesis architecture combining tissue-guided bottleneck modulation, prior-restoration fusion, and decoder refinement to preserve white-matter structure. Second, we evaluate synthetic FA using image-level metrics, FA-stratified white-matter assessment, and JHU atlas-based tract-level biomarker preservation. Experimental results on the HCP Young Adult dataset show that ATLANTIS achieves consistent image-level improvements and larger relative gains in tract-level biomarker preservation, suggesting that anatomy-guided synthesis is most clearly reflected when evaluation moves from voxel-level similarity to tract-level biomarkers.

II. Related Work

A. Structural-to-Diffusion MRI Synthesis

Medical image synthesis has become an active area of research for estimating missing or alternative imaging modalities, with recent reviews summarizing progress across MRI, CT, and PET synthesis using CNN-, Transformer-, and diffusion-based models [5]. Pix$_2$Pix [6] established conditional GANs as a standard framework for paired image-to-image translation. CNN-based architectures such as U-Net [15] and 3D U-Net [7] enabled direct regression-based synthesis with multi-scale feature learning. More recently, ResViT [8] combined residual convolutional blocks with vision transformers for multimodal MRI synthesis, and I^{2}SB [9] introduced diffusion-bridge models that learn nonlinear transport between image distributions. These methods form the baseline categories against which we evaluate ATLANTIS.

Directly relevant to our work are methods that specifically target T1w-to-FA or T1w-to-diffusion scalar map synthesis. Gu et al. [3] demonstrated that CycleGAN can generate FA and mean diffusivity maps from T1w images. Anctil-Robitaille et al. [10] proposed a manifold-aware CycleGAN operating on the Riemannian manifold of diffusion tensors, making it one of the few studies to include tractography-based evaluation of synthetic diffusion outputs. Murcia Tapias et al. [11] used a simplified GAN with reduced parameters for T1w-to-FA synthesis, and Macro2Micro [12] recently introduced multi-scale structural brain encoding for T1w-to-FA translation. Kwon et al. [13] further investigated generative T1-to-FA synthesis with transfer learning for white-matter assessment in stroke cohorts. Beyond T1w inputs, Chan et al. [16] synthesized FA and mean diffusivity maps from FLAIR MRI and included tissue-level regional analyses, indicating increasing interest in evaluating synthetic diffusion scalar maps beyond global image similarity. Recent studies have further examined T1w-derived diffusion scalar maps in clinical-utility settings, including FA/MD synthesis for dementia prediction in cerebral small vessel disease [17] and reversible-GAN-based synthesis of DTI scalar maps [18]. These works collectively establish the feasibility of structural-to-diffusion synthesis. However, for DWI-derived white-matter anisotropy synthesis, systematic atlas-based evaluation of tract-level FA preservation remains underexplored.

B. Biomarker-Oriented Validation of Synthetic FA

In population neuroimaging and clinical studies, FA is typically quantified as tract-mean values within anatomically defined white-matter regions rather than as per-voxel intensities. Unlike voxel-wise similarity metrics, tract-level FA statistics aggregate predictions over anatomically meaningful white-matter pathways and directly correspond to quantities used in downstream group-level or clinical analyses. The JHU ICBM-DTI-81 white-matter atlas [19] and associated tract probability maps [20] provide a standardized parcellation of major white-matter tracts. Atlas-based whole-brain white-matter analysis has been applied in population and disease studies, from early demonstrations in aging and Alzheimer's disease cohorts [21] to recent large-scale investigations in over 30,000 UK Biobank participants [22], demonstrating tract-level FA quantification as a practical and scalable biomarker analysis strategy. Multi-site ENIGMA-DTI studies further demonstrate coordinated cross-cohort analysis of diffusion-derived FA phenotypes [23].

For synthetic medical images, reference-based voxel metrics such as SSIM and PSNR can be insufficient to capture clinical or biomarker utility. Dohmen et al. [14] showed that these metrics are sensitive to intensity normalization, systematically underestimate blurring, and correlate poorly with downstream task performance. These findings motivate a more targeted evaluation of synthetic FA that directly assesses whether tract-level FA values are preserved. In this work, we evaluate synthetic FA using JHU atlas-based tract-level MAE, Pearson correlation, Spearman correlation, and absolute bias, directly assessing whether synthetic FA preserves the tract-level quantities used as white-matter biomarkers.

III. Method

A. Problem Formulation

Given a T1w 2.5D slice stack $\mathbf{T} \in \mathbb{R}^{5\times H\times W}$ (five adjacent axial slices, z-score normalized), white matter (WM) and gray matter (GM) tissue probability maps $\mathbf{P} \in \mathbb{R}^{2\times H\times W}$ derived from the same T1w images via FastSurfer, and a binary brain mask $\mathbf{M} \in \{0,1\}^{H\times W}$, we concatenate them into an 8-channel input $\mathbf{X} = [\mathbf{T}; \mathbf{P}; \mathbf{M}]$. ATLANTIS learns a mapping $f_\theta : \mathbf{X} \to \hat{\mathbf{Y}}$ that predicts an FA map $\hat{\mathbf{Y}} \in [0,1]^{H\times W}$. Tissue priors are derived from T1w MRI alone, requiring no diffusion acquisition. Here, FA serves as the target DWI/DTI-derived white-matter anisotropy biomarker.

B. Overview of ATLANTIS

Figure 1 illustrates the ATLANTIS architecture. The input $\mathbf{X}$ is first processed by an Anatomy Encoder, a U-Net-style multi-scale encoder that extracts hierarchical anatomical features. A Tissue Query Adapter (TQA) then modulates the encoder bottleneck via cross-attention, using the WM/GM tissue probability maps as spatial queries. From the tissue-modulated bottleneck, three parallel branches emerge: a Prior Head that predicts a coarse FA estimate, a Beta Head that produces voxel-wise fusion weights, and a Residual Latent Bridge that provides training-only regularization. The Prior Head output is refined by a Restoration Decoder, which fuses multi-scale anatomical skip connections from the Anatomy Encoder to recover fine structural detail. Finally, the prior and restored FA predictions are combined through voxel-wise adaptive fusion: $\hat{\mathbf{Y}} = \boldsymbol{\beta} \odot \mathbf{Y}_{\text{prior}} + (1 - \boldsymbol{\beta}) \odot \mathbf{Y}_{\text{restored}}$.

Unless otherwise specified, ATLANTIS denotes the model after decoder refinement. The pre-refinement version, used only in ablation comparisons, is referred to as Base ATLANTIS.

C. Anatomy Encoder and Tissue Query Adapter

The Anatomy Encoder follows a 4-level U-Net encoder design. It maps the 8-channel input through successive down-sampling stages, producing multi-scale feature maps $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{E}_4\}$ (from fine to coarse) and a bottleneck feature $\mathbf{B} \in \mathbb{R}^{C_b \times h \times w}$. The encoder skip features $\mathbf{E}_1$–$\mathbf{E}_4$ serve as anatomical priors for the Restoration Decoder, providing spatial context at multiple resolutions.

The Tissue Query Adapter injects tissue-specific information into the bottleneck through multi-head cross-attention using a standard query-key-value formulation [24]. The tissue prior $\mathbf{P}$ is first downsampled to the bottleneck spatial resolution $(h \times w)$, then projected into query tokens $\mathbf{Q}_T$ via a convolutional stem. The bottleneck feature $\mathbf{B}$ is projected into key and value tokens $\mathbf{K}_B$ and $\mathbf{V}_B$ through 1×1 convolutions:

$$\mathbf{B}' = \mathbf{B} + \text{Attn}(\mathbf{Q}_T, \mathbf{K}_B, \mathbf{V}_B). \tag{1}$$

The residual connection preserves the original anatomical information while the attention output selectively enhances tissue-relevant features. This design aims to provide tissue-aware bottleneck modulation beyond simple input-level concatenation of tissue channels.

D. Prior-Restoration Adaptive Fusion

The Prior Head takes the tissue-modulated bottleneck $\mathbf{B}'$ and up-samples it through four transposed convolution stages to produce a coarse FA mean $\boldsymbol{\mu}_{\text{prior}} \in [0,1]^{H \times W}$ and an auxiliary log-variance map $\mathbf{s}_{\text{prior}}$. The log-variance map provides an uncertainty-aware prior representation and is concatenated with the prior mean to form the input to the Restoration Decoder.

The Restoration Decoder refines this coarse prior. It takes $[\boldsymbol{\mu}_{\text{prior}}, \mathbf{s}_{\text{prior}}]$ as a two-channel input and processes it through a 4-level encoder-decoder structure. At each encoder level, the corresponding Anatomy Encoder skip feature $\mathbf{E}_i$ is concatenated with the decoder features, providing multi-scale anatomical guidance. The decoder outputs a restored FA mean $\boldsymbol{\mu}_{\text{restored}} \in [0,1]^{H \times W}$ and a restored log-variance map $\mathbf{s}_{\text{restored}}$.

The Beta Head shares the same bottleneck $\mathbf{B}'$ as the Prior Head and predicts a voxel-wise fusion weight map $\boldsymbol{\beta} \in [0,1]^{H \times W}$ through a symmetric up-sampling pathway. The final FA prediction is obtained by adaptive fusion of the prior and restored means:

$$\hat{\mathbf{Y}} = \boldsymbol{\beta} \odot \boldsymbol{\mu}_{\text{prior}} + (1 - \boldsymbol{\beta}) \odot \boldsymbol{\mu}_{\text{restored}}. \tag{2}$$

Only the FA means are fused; the restored log-variance $\mathbf{s}_{\text{restored}}$ is used directly in the heteroscedastic reconstruction loss (Section III-F). This formulation allows the model to learn per-voxel preferences: trusting the prior where anatomy strongly constrains FA, and relying on restoration where finer detail is needed.

E. Residual Latent Regularization

A frozen FA autoencoder, pre-trained on training-set real FA maps, provides a 32-dimensional latent space. The Residual Latent Bridge maps the spatially pooled bottleneck $\mathbf{B}'$ through an MLP to a latent residual vector of the same dimensionality. During training, an L2 penalty is applied to this residual:

$$\mathcal{L}_{\text{bridge}} = \| \text{Bridge}(\mathbf{B}') \|_2^2. \tag{3}$$

This term acts as a training-only regularizer that discourages the bottleneck representation from deviating into regions that would require large latent corrections, without enforcing explicit alignment to specific FA latent codes. The Bridge and FA autoencoder are removed at inference, introducing no additional computation.

F. Training Objective

The model is trained by minimizing a composite loss applied to the fused FA output $\hat{\mathbf{Y}}$. The primary terms are a masked L1 reconstruction loss and a heteroscedastic Gaussian negative log-likelihood (NLL), following input-dependent uncertainty modeling [25], that jointly optimizes the predicted mean and log-variance:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2|\Omega|} \sum_{\Omega} \left(\mathbf{s}_{\text{restored}} + (\hat{\mathbf{Y}} - \mathbf{Y})^2 \odot \exp(-\mathbf{s}_{\text{restored}}) \right), \tag{4}$$

where Ω denotes the set of brain-masked voxels and $\mathbf{s}_{\text{restored}}$ is the log-variance map from the Restoration Decoder. Quality terms encourage structural and statistical consistency: SSIM, spatial gradient L1, per-sample distribution matching (mean, standard deviation, quantiles), and per-sample Pearson correlation. Regularization terms include an inter-subject WM-mean contrast loss (variance and covariance), an L1 consistency term between prior and restored outputs, and the bridge L2 penalty (Section III-E). All loss components are computed within the brain mask.

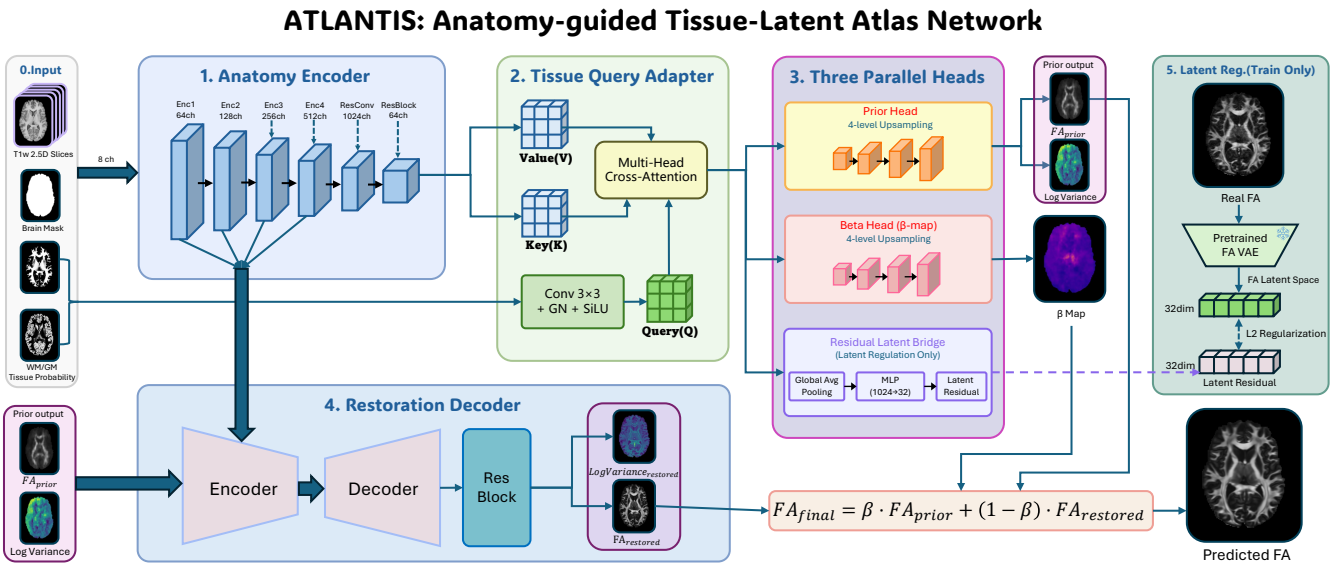


Fig. 1. Overview of the ATLANTIS architecture. T1w 2.5D slices, WM/GM tissue priors, and brain mask are concatenated as input. The Anatomy Encoder extracts multi-scale features; TQA injects tissue-guided cross-attention into the bottleneck. Prior Head predicts a coarse FA estimate; the Restoration Decoder refines it using anatomy skip connections. Beta Head produces voxel-wise β for adaptive fusion of prior and restored FA. A frozen FA autoencoder provides training-only latent regularization via the Residual Latent Bridge.

G. Decoder Refinement

After the main training stage, we perform a lightweight decoder refinement phase. The Anatomy Encoder, TQA, Prior Head, and FA autoencoder are frozen, and only the Restoration Decoder and Beta Head are fine-tuned with a reduced learning rate under the same loss. This stage targets improved tract-level FA calibration without altering the anatomical feature extraction that underlies the prior prediction, and the resulting model is designated as ATLANTIS.

IV. Experiments and Results

A. Dataset, Baselines, and Evaluation Protocol

We evaluate on the Human Connectome Project (HCP) Young Adult dataset, where T1w and diffusion-weighted images are pre-aligned to a common space. The input consists of a 2.5D T1w slice stack (five adjacent axial slices, z-score normalized), WM and GM tissue probability maps obtained via FastSurfer [26], and a binary brain mask, concatenated to form an 8-channel input. The target is the FA map clipped to $[0, 1]$. We adopt an approximately 80/10/10 train/validation/test split. All splits were performed at the subject level to avoid slice-level leakage. One subject with severe misregistration is excluded, leaving 97 test subjects. All voxel-wise metrics were computed within the brain mask.

Five representative baselines are compared across four methodological categories: Pix_2Pix (GAN-based paired translation, 57M parameters), 3D U-Net (volumetric CNN regression, 6.2M), I^2SB (diffusion-bridge-based translation, 30M), 2.5D U-Net (supervised 2.5D CNN regression, 28M), and ResViT (Transformer-based translation, 123M). ATLANTIS has 108M parameters. All baselines were trained with T1w input; ATLANTIS additionally uses WM/GM tissue priors derived from the same T1w images via FastSurfer, requiring no additional diffusion information. Because ATLANTIS uses additional T1w-derived prior channels, we performed an input-control experiment by adding the same WM/GM priors and brain mask to the 2.5D U-Net. This augmented baseline achieved MAE 0.0438 ± 0.0038, comparable to the T1w-only 2.5D U-Net (0.0436 ± 0.0036; paired t-test $p > 0.05$), suggesting that ATLANTIS gains arise from explicit tissue-aware bottleneck modulation rather than naive channel concatenation. Unless otherwise noted, ATLANTIS refers to the model after decoder refinement; Base ATLANTIS denotes the pre-refinement version used only in ablation comparisons.

Evaluation is conducted at three levels. First, image-level metrics include mean absolute error (MAE), root mean square error (RMSE), Pearson correlation, Spearman correlation, and peak signal-to-noise ratio (PSNR). Voxel-wise Pearson and Spearman correlations were computed per subject within the brain mask and then averaged across test subjects. Second, FA-stratified metrics assess performance within specific FA ranges: WM-Pearson is computed over voxels with FA $>$ 0.2, and Deep-WM MAE over voxels with FA > 0.35. These ranges serve as evaluation strata and should not be interpreted as anatomical tissue segmentation. Third, tract-level evaluation quantifies biomarker preservation using the JHU ICBM-DTI-81 white-matter atlas. We extract mean FA within each of eight major tracts with left and right hemispheres merged: Cingulum (CG), Corona Radiata (CR), Corpus Callosum (CC), Inferior Fronto-Occipital Fasciculus (IFOF), Inferior Longitudinal Fasciculus (ILF), Internal Capsule (IC), Superior Longitudinal Fasciculus (SLF), and Uncinate Fasciculus (UF). For

each tract, Pearson and Spearman correlations were computed across the 97 test subjects using per-subject tract-mean FA, and then averaged over the eight tracts, yielding four aggregate metrics: average tract-level MAE, average tract-level Pearson, average tract-level Spearman, and average absolute bias.

B. Image-Level FA Synthesis Benchmark

Table I reports the image-level benchmark across all six methods on 97 test subjects. ATLANTIS achieves the best performance among all evaluated baselines on every metric. Specifically, ATLANTIS attains an MAE of 0.0402 and Pearson correlation of 0.9484, reducing MAE by 5.0% relative to ResViT (MAE 0.0423) and by 7.8% relative to 2.5D U-Net (MAE 0.0436). ATLANTIS also achieves the best RMSE (0.0612), Spearman correlation (0.9298), PSNR (24.21 dB), and GM-Pearson (0.6264) among the evaluated baselines, consistent with its overall image-level advantage. In a paired t-test with Bonferroni correction for the eight comparisons in Table I, all differences between ATLANTIS and ResViT were significant (MAE, RMSE, Pearson, Spearman, PSNR, WM-Pearson, Deep-WM MAE: all $p < 10^{-3}$; GM-Pearson: $p = 0.003$).

On FA-stratified metrics, ATLANTIS achieves a WM-Pearson of 0.8923 and Deep-WM MAE of 0.0580, compared with 0.8829 and 0.0619 for ResViT, respectively. The relative gain is slightly larger in Deep-WM MAE (6.3%) than in global MAE (5.0%), suggesting that ATLANTIS may be particularly beneficial in high-FA regions.

C. Tract-Level White-Matter Biomarker Preservation

Table II reports tract-level biomarker preservation aggregated across eight JHU white-matter tracts. This is the central evaluation: tract-level metrics assess whether synthetic FA preserves DWI/DTI-derived biomarker properties beyond voxel-wise accuracy.

ATLANTIS achieves the best results on all four tract-level metrics. It attains an average tract MAE of 0.0067, average tract Pearson of 0.9135, average tract Spearman of 0.9046, and average absolute bias of 0.0008. Compared with ResViT (tract MAE 0.0084, tract Pearson 0.8999, tract Spearman 0.8876, abs bias 0.0048), ATLANTIS reduces average tract MAE by 20.2% and average absolute bias by 83.3%. The near-zero absolute bias (0.0008) indicates that ATLANTIS substantially reduces systematic tract-level deviation, a property that is important for tract-based FA quantification.

The tract-level advantage of ATLANTIS is substantially larger than its image-level advantage: compared with ResViT, global MAE improves by 5.0%, whereas tract MAE improves by 20.2% and absolute bias decreases by 83.3%. This increasing gain from global voxels to tract-level metrics suggests that ATLANTIS improves biomarker preservation rather than only voxel-wise fidelity.

Comparing Base ATLANTIS (tract MAE 0.0070, tract Pearson 0.9112) with ATLANTIS (tract MAE 0.0067, tract Pearson 0.9135) shows that decoder refinement contributes a further 4.5% tract MAE reduction, confirming its targeted benefit for tract-level calibration. I^{2}SB further illustrates this mismatch: despite reasonable image-level performance, its tract-level error remains high (tract MAE 0.0184, abs bias 0.0178).

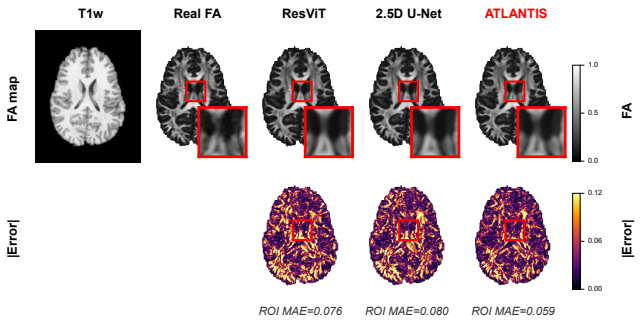


Fig. 2. Qualitative FA synthesis results for a representative test slice. ROI insets show absolute error maps for the highlighted white-matter region.

Figure 3 provides a tract-wise breakdown of MAE across the eight JHU tracts, showing that ATLANTIS achieves the lowest per-tract MAE on the majority of tracts, with particularly clear advantages in the Corona Radiata and Internal Capsule.

D. Qualitative Comparison

Figure 2 presents a qualitative comparison of T1w input, real FA, and predicted FA maps from ResViT, 2.5D U-Net, and ATLANTIS for a representative test slice. ROI insets highlight the same high-FA white-matter region across all models, with the inset showing absolute error ($|\text{prediction} - \text{real FA}|$, range $[0, 0.12]$) to reveal local differences that are invisible at full-FOV scale.

All three methods preserve global brain anatomy, and their predicted FA maps are visually similar at full-FOV resolution. The ROI insets reveal that ATLANTIS produces visibly lower local error in the highlighted high-FA region compared with ResViT and 2.5D U-Net, consistent with the quantitative Deep-WM MAE improvement reported in Table I.

E. Ablation Study

Table III reports the ablation study examining the contribution of each architectural component.

Seven variants are compared: (1) removing the Restoration Decoder and Beta Head, so the final output is the Prior Head prediction alone (w/o Restoration Decoder); (2) removing tissue prior channels from the input and disabling TQA (w/o Tissue Prior); (3) replacing TQA with an identity mapping while retaining tissue channels in the input (w/o TQA); (4) removing the Residual Latent Bridge regularization (w/o Latent Bridge); (5) removing the Beta Head and using the restored output directly without adaptive fusion (w/o Adaptive Fusion); (6) the complete framework before decoder refinement (Base ATLANTIS); and (7) the full ATLANTIS after refinement.

Restoration Decoder is the most critical component. Removing the Restoration Decoder and Beta Head increases

TABLE I
IMAGE-LEVEL FA SYNTHESIS BENCHMARK (97 TEST SUBJECTS, MEAN ± STD ACROSS SUBJECTS).

Method	MAE ↓	RMSE ↓	Pearson ↑	Spearman ↑	PSNR ↑	WM-Pearson ↑	GM-Pearson ↑	Deep-WM MAE ↓
Pix_2Pix	0.0735±0.0032	0.1109±0.0045	0.8299±0.0143	0.7858±0.0172	19.10±0.35	0.7326±0.0271	0.2671±0.0346	0.1021±0.0063
3D U-Net	0.0578±0.0023	0.0826±0.0035	0.9060±0.0089	0.8707±0.0113	21.66±0.36	0.8273±0.0187	0.4351±0.0262	0.0856±0.0071
I^2SB	0.0538±0.0048	0.0788±0.0074	0.9148±0.0232	0.8856±0.0224	22.07±0.76	0.8361±0.0376	0.4697±0.0402	0.0761±0.0095
2.5D U-Net	0.0436±0.0036	0.0662±0.0053	0.9385±0.0122	0.9212±0.0144	23.58±0.66	0.8709±0.0245	0.6112±0.0404	0.0635±0.0054
ResViT	0.0423±0.0038	0.0639±0.0053	0.9432±0.0117	0.9254±0.0143	23.89±0.68	0.8829±0.0231	0.6227±0.0406	0.0619±0.0057
ATLANTIS	**0.0402±0.0041**	**0.0612±0.0057**	**0.9484±0.0122**	**0.9298±0.0151**	**24.21±0.75**	**0.8923±0.0237**	**0.6264±0.0421**	**0.0580±0.0059**

TABLE II
JHU TRACT-LEVEL BIOMARKER PRESERVATION (97 SUBJECTS, 8 TRACTS; AVG TRACT MAE REPORTED AS MEAN ± STD ACROSS SUBJECTS).

Method	Avg tract MAE ↓	Avg tract Pearson ↑	Avg tract Spearman ↑	Avg abs bias ↓
I^2SB	0.0184±0.0061	0.9065	0.8954	0.0178
ResViT	0.0084±0.0042	0.8999	0.8876	0.0048
2.5D U-Net	0.0075±0.0038	0.9060	0.8933	0.0025
Base ATLANTIS	0.0070±0.0036	0.9112	0.9008	0.0018
ATLANTIS	**0.0067±0.0035**	**0.9135**	**0.9046**	**0.0008**

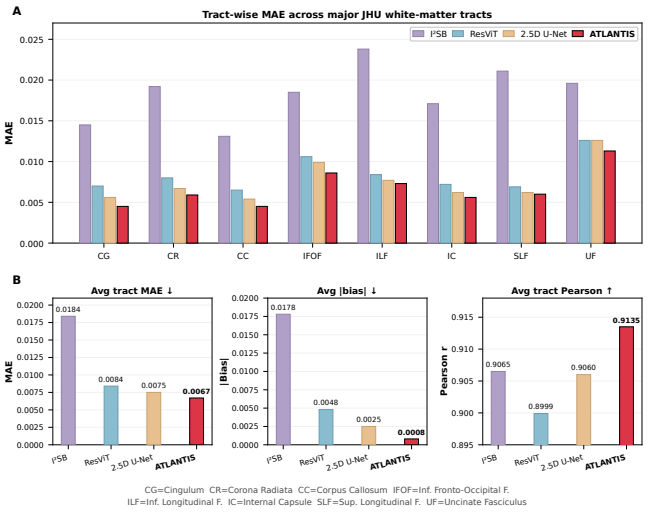


Fig. 3. Tract-wise MAE across eight JHU white-matter tracts. ATLANTIS achieves the lowest per-tract MAE on the majority of tracts.

TABLE III
ABLATION STUDY ON ARCHITECTURAL COMPONENTS.

Variant	MAE ↓	Pearson ↑	WM-Pearson ↑	Deep-WM MAE ↓
w/o Restoration Decoder	0.0544	0.9108	0.8264	0.0819
w/o Tissue Prior	0.0422	0.9436	0.8876	0.0604
w/o TQA	0.0422	0.9436	0.8875	0.0592
w/o Latent Bridge	0.0421	0.9435	0.8871	0.0603
w/o Adaptive Fusion	0.0420	0.9433	0.8859	0.0597
Base ATLANTIS	0.0412	0.9454	0.8926	0.0581
ATLANTIS	**0.0402**	**0.9484**	**0.8923**	**0.0580**

MAE from 0.0402 to 0.0544 (a 35.3% relative increase), decreases WM-Pearson from 0.8923 to 0.8264, and raises Deep-WM MAE from 0.0580 to 0.0819 (a 41.2% increase). The magnitude of degradation confirms that multi-scale anatomy-guided restoration is essential for reconstructing fine white-matter detail.

Tissue Query Adapter provides consistent but moderate improvement. Both w/o Tissue Prior and w/o TQA yield MAE 0.0422, compared with 0.0412 for Base ATLANTIS, indicating that naive tissue-channel concatenation is insufficient. Explicit tissue-guided bottleneck modulation contributes approximately 2.4% MAE reduction and provides the main benefit of the T1w-derived tissue priors.

Latent regularization and adaptive fusion provide complementary but smaller gains. Removing the Residual Latent Bridge or Beta Head slightly degrades MAE from 0.0412 to 0.0421 and 0.0420, respectively, with similar decreases in WM-Pearson, indicating that both components support the full framework.

Decoder refinement improves global accuracy and tract-level calibration. Compared with Base ATLANTIS, ATLANTIS reduces MAE from 0.0412 to 0.0402 and tract MAE from 0.0070 to 0.0067, while keeping WM-Pearson essentially unchanged.

V. DISCUSSION AND CONCLUSION

We presented ATLANTIS, an anatomy-guided tissue-latent atlas network for DWI-derived white-matter anisotropy synthesis from T1w MRI, instantiated here as FA map synthesis. ATLANTIS achieved strong image-level performance among the evaluated baselines, reducing MAE by 5.0% relative to ResViT. More importantly, its advantage was larger at the biomarker level, reducing average tract MAE by 20.2% and average absolute bias by 83.3% across eight JHU white-matter tracts. These results support the central premise of this work: voxel-level fidelity alone is insufficient for assessing synthetic FA intended for tract-based biomarker analysis.

The disproportionate tract-level gains show that anatomy-guided synthesis preserves tract-mean FA beyond what global metrics reveal. Although all methods produced visually plausible FA maps, their tract-level errors and biases differed substantially, reinforcing the need for atlas-based biomarker evaluation in synthetic FA studies [14], [19], [21], [22].

Ablation results further clarify the role of the proposed components. The Restoration Decoder is the most influential module, indicating that multi-scale anatomy-guided refinement is important for recovering white-matter detail. The Tissue Query Adapter and tissue prior provide smaller but consistent improvements, suggesting that explicit tissue-aware bottleneck modulation is more effective than relying only on input-level tissue concatenation. Decoder refinement further improves global accuracy and tract-level calibration.

Several limitations remain. Evaluation was limited to the HCP Young Adult cohort, and generalization to other scanners, protocols, age ranges, and clinical populations remains to be established. Tract-level analysis used eight left-right-merged JHU tracts, without finer parcellations, tract profiles, or tractography-based analyses. This study focuses on FA as a representative DWI-derived white-matter anisotropy biomarker; extending ATLANTIS to other diffusion scalar maps or tensor-level representations remains future work. The current study also lacks downstream validation against cognitive, demographic, or clinical endpoints, and tissue-prior quality may vary across datasets. Future work should address multi-site validation, broader tract coverage, uncertainty calibration, and task-based evaluation.

In summary, ATLANTIS demonstrates that anatomy-guided synthesis of DWI-derived white-matter anisotropy from T1w MRI can improve both image-level FA fidelity and tract-level biomarker preservation. Systematic atlas-based tract evaluation provides a complementary criterion for assessing synthetic FA quality beyond voxel-wise similarity and may be useful for future structural-to-diffusion MRI synthesis studies.

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